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PRELIMINARY RESULTS OF A STUDY OF THE PALEOMAGNETISM OF THE SUDBURY IRRUPTIVE

(Report and 5 figures)

A. Larochelle

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DEPARTMENT OF ENERGY, MINES AND RESOURCES

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ABSTRACT

Measurements of paleomagnetic directions within the Sudbury irruptive were obtained from 265 cores drilled at 74 norite sites and 197 cores drilled at 55 micropegmatite sites. After elimination of data considered unreliable according to preset criteria of acceptance, the remaining directions of magnetization (following a.f. cleaning in a peak field of 250 oe) were analyzed quantitatively and in terms of the known geological context. The low angular dispersion of the magnetization directions within large sectors of the irruptive and their remarkably high stability during progressive demagnetization suggest that the magnetization measured is predominantly primary and of thermoremanent origin. This assumption was provisionally made in interpreting the results.

A statistical analysis shows that the mean directions of magnetization of the norite and of the micropegmatite are not significantly different in undisturbed sectors of the irruptive. It is inferred from this that the two rock units are probably contemporaneous. The low angular dispersion of the site mean directions of magnetization along profiles across the irruptive suggests further that either the irruptive cooled rather rapidly or the paleomagnetic field was more stable at that time than at present.

The data reported in this study were combined with data previously reported for the paleomagnetism of the irruptive, the blending of the two sets of data having been justified on a statistical basis. The previously published geological interpretations of the irruptive's paleomagnetism still hold their respective merits although their original basis is more firmly established as a result of the present study. The validity of these interpretations can only be tested through considerably more work which this study should help in planning.

PRELIMINARY RESULTS OF A STUDY OF THE PALEOMAGNETISM OF THE SUDBURY IRRUPTIVE

INTRODUCTION

The Sudbury Irruptive has inspired a voluminous body of geological literature and yet its mode of origin still remains enigmatic from many points of view. Figure 1, a simplified geological map, shows that the irruptive has a roughly oval shape composed of an inner member of micropegmatite and an outer layer of a basic rock commonly referred to as norite. The rocks of both members are known to dip concordantly toward the centre of the structure and they are presently believed to be about 1700 m.y. old (Fairbairn et al., 1968).

Previous studies on the paleomagnetism of the irruptive by Hood (1961) and Sopher (1963) pointed out that the mean directions of magnetization in the south and north sections of the oval diverge appreciably from one another. This observation suggested to both authors that the present shape of the irruptive is partly due to post-irruptive tectonic activity. Sopher had also noted that the paleomagnetic directions obtained from the norite are considerably more consistent than those obtained from the micropegmatite but he did not make a quantitative appraisal of this observation. Finally, he noted that within sectors of the irruptive the directions of magnetization of the micropegmatite and the norite were nearly parallel whereas they diverged appreciably from one another in other sectors. As it was difficult for him to 'filter' his data prior to analysis (on account of the sampling scheme generally adopted at the time he did his work), he could not easily explain this paradox on a quantitative basis.

From the early phase of the present study the sampling scheme had two main objectives: (1) concentrated sampling carried out in undisturbed sectors of the irruptive to establish the relative paleomagnetic 'reliability' of the micropegmatite and norite rock types, to evaluate their age relationship and to assess the importance of the paleomagnetic field secular variation, and (2) subject to outcrop distribution and accessibility, sampling carried out throughout each of the north, the south and the northeast portions of the irruptive to verify on a more rigorous statistical basis the findings of the earlier paleomagnetic studies.

To obtain the data reported below, improved sampling, measuring and testing techniques were used, a more comprehensive sampling scheme was designed, and rigid criteria were adopted for the acceptance of data as suitable for a quantitative interpretation. The data obtained so far are not sufficiently abundant to permit a comprehensive interpretation in geological terms but they provide a firm basis for study of a paleomagnetic picture which now appears to be more complex than originally envisaged.

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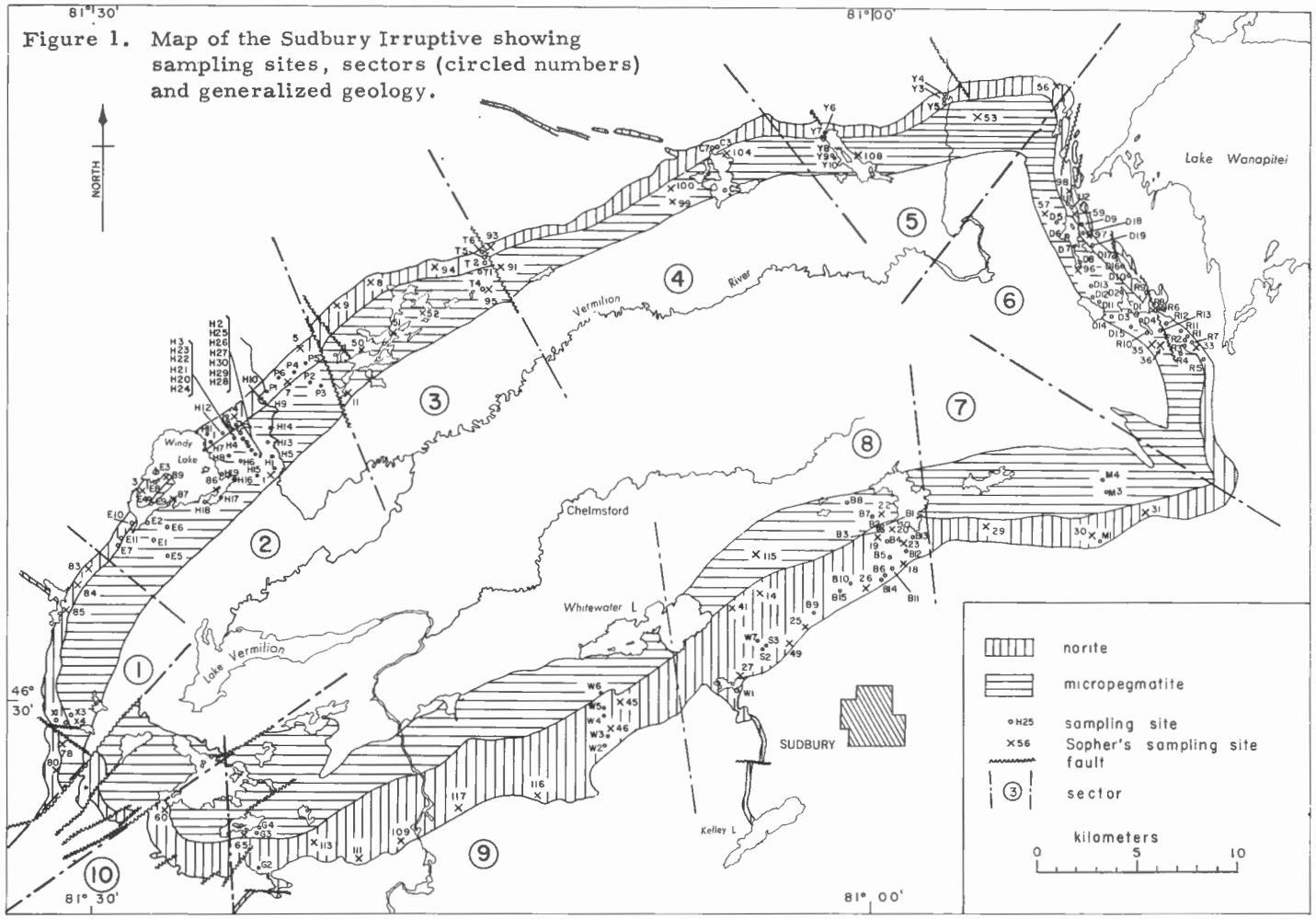


Figure 1. Map of the Sudbury Irruptive showing sampling sites, sectors (circled numbers) and generalized geology.

For a statistical analysis of these data it was found convenient to sub-divide the Sudbury Irruptive into ten sectors (Fig. 1) whose boundaries are either fault zones or major swings in the trend of the irruptive rocks. Within each sector the data obtained for the norite and the micropegmatite were analyzed separately. The theoretical basis of this analysis will not be given in detail below and the reader is referred elsewhere for a complete description and for definitions of the nomenclature used (Larochelle, 1968).

Acknowledgments

This work is part of a joint project undertaken by the International Nickel Company of Canada, Ltd. and the Geological Survey of Canada. Mr. S.R. Sopher and Dr. J. Guy Bray of the International Nickel Company selected the sampling sites and supervised the field work. The laboratory measurements were done at the Geological Survey of Canada. The manuscript was critically and helpfully reviewed by Dr. Bray and Mr. Sopher, although the writer assumes final responsibility for the interpretation of the data.

SAMPLING AND MEASUREMENTS

At each sampling site (Fig. 1) two to five cores were drilled between three to five metres apart and oriented in situ with a solar compass. In all, 265 cores of norite were drilled at 74 sites and 197 cores of micropegmatite were drilled at 55 sites. From each core, two cylinders were cut and the natural remanent magnetization (NRM) of each cylinder was measured with a biastatic magnetometer whose angular limit of resolution is 0.5 degree or less (Larochelle and Christie, 1967). The cylinders from sites marked with an asterisk in Tables 1a, 1b were 2.5 cm in diameter and 2.2 cm in height; all other cylinders were 3.2 cm in diameter by 2.8 cm in height.

MAGNETIZATION STABILITY TEST

At least one cylinder from each site was exposed to successive alternating field treatments of stepwise increasing intensity with the apparatus described by Larochelle and Black (1965). Demagnetization curves (Figs. 2a, 2b) and stereographic projections of the successive directions of magnetization (Figs. 3a, 3b) for typical samples of each rock type illustrate that the magnetization of the norite is in general considerably more stable than that of the micropegmatite. For both rock types however it appears that the component of viscous magnetization may be eliminated by magnetic cleaning in a demagnetizing field of 250 oe (peak). As a result of cleaning the 924 cylinders in a demagnetizing field of this intensity the magnetization of 12 norite and 14 micropegmatite cores (52 cylinders) became too weak to be accurately measured.

Table 1a: Norite Site Mean Directions

S	s	N	n	D (deg)	I (deg)	$\delta_{\bar{R}}$ (deg)	S	s	N	n	D (deg)	I (deg)	$\delta_{\bar{R}}$ (deg)
1	*X1	3	2	234.0	24.0	15.1	6	D16	4	2	63.0	43.4	5.6
1	*X4	2	2	236.5	54.5	2.5	6	D17	4	3	69.8	-30.7	6.4
2	E3	4	4	295.3	60.3	1.6	6	D18	4	4	70.5	31.2	1.8
2	E4	4	4	289.7	68.6	1.6	6	D19	4	4	65.2	32.3	2.3
2	E7	4	4	319.0	64.1	7.5	6	*R1	3	2	76.5	48.1	0.4
2	E8	4	4	285.6	62.0	1.5	6	*R2	3	2	20.7	62.5	5.0
2	E10	4	2	245.2	54.7	2.3	6	*R3	3	3	300.1	65.8	5.1
2	E11	3	2	282.2	38.7	15.0	6	R5	4	3	58.7	48.3	5.4
2	*H2	3	3	279.9	61.0	5.3	6	R6	4	2	56.1	60.2	1.8
2	*H3	3	3	285.2	63.0	0.6	6	R7	4	4	62.7	46.6	2.7
2	H7	5	5	304.9	67.3	0.6	6	R8	4	4	51.8	51.6	4.0
2	H9	4	4	272.1	62.7	1.0	6	R9	4	1	238.8	-01.3	-
2	H10	4	4	294.4	65.2	0.9	6	R11	4	4	62.0	42.4	1.4
2	H11	4	4	291.5	65.8	0.3	6	R12	4	4	48.0	49.9	1.3
2	H12	4	4	292.6	64.8	1.5	7	*M1	3	3	193.8	45.1	8.0
2	H20	4	4	305.5	61.9	1.1	8	*B1	3	0	-	-	-
2	H21	4	4	301.8	63.7	0.8	8	*B2	3	2	149.6	61.8	1.8
2	H22	4	4	302.9	66.2	1.0	8	*B3	3	2	198.5	82.4	8.6
2	H23	4	4	316.2	65.0	1.1	8	*B4	3	3	159.0	44.8	5.7
2	H24	4	1	284.1	55.0	-	8	*B5	3	3	168.2	59.0	0.8
2	P1	4	4	298.7	65.7	2.0	8	*B6	3	3	178.2	65.6	2.1
2	P4	4	4	274.2	60.5	2.5	8	B9	4	4	163.5	58.6	1.3
2	P5	5	5	284.1	69.1	2.2	8	B10	4	4	163.6	63.4	1.7
2	P6	4	4	298.7	64.0	0.7	8	B11	4	4	187.7	70.5	1.3
3	*T2	3	2	317.9	60.7	9.3	8	B12	4	4	171.0	63.2	0.9
3	*T5	3	2	260.4	66.3	2.4	8	B13	4	4	178.6	61.8	1.1
3	*T6	3	3	308.9	74.9	4.6	8	B14	4	4	173.6	68.9	1.5
4	*C3	3	2	276.7	09.0	64.0	8	B15	4	4	177.3	61.9	1.5
4	*C7	3	1	024.5	-01.8	-	8	*W1	3	3	199.4	65.8	1.2
5	*Y3	2	0	-	-	-	8	W7	4	4	176.0	68.6	3.2
5	*Y4	2	2	348.6	73.5	3.0	8	S2	4	4	176.0	59.9	4.3
5	*Y5	2	0	-	-	-	8	S3	4	4	169.0	60.2	1.3
5	*Y6	2	1	348.6	-22.6	-	9	*W2	3	3	149.8	57.8	2.7
5	*Y7	2	2	190.4	72.9	9.8	9	*W3	3	3	150.8	53.2	0.9
6	U1	4	0	-	-	-	9	*W4	3	3	190.7	59.0	4.1
6	U2	4	3	076.1	30.2	4.7	9	*W5	3	3	188.3	75.1	1.5
6	D9	4	2	086.9	74.5	5.8	9	*G2	3	1	337.8	36.2	-
6	D10	4	3	059.8	38.4	2.2							

S: sector number; s, site number; N, total number of cores collected; n, number of cores accepted as reliable; D, I, site mean declination and inclination; $\delta_{\bar{R}}$, standard error of site mean. Only the data listed opposite underlined site numbers were considered significant.
* Cylinders from these sites were 2.5 cm by 2.2 cm – others were 3.2 cm by 2.8 cm.

Table 1b: Micropegmatite Site Mean Directions

(for explanation of Symbols see Table 1a)

S	s	N	n	D (deg)	I (deg)	$\delta_{\bar{R}}$ (deg)	S	s	N	n	D (deg)	I (deg)	$\delta_{\bar{R}}$ (deg)
1	*X3	3	2	240.3	46.2	6.6	4	*C5	2	0	-	-	-
2	E1	4	4	265.8	61.8	36.0	5	*Y8	2	0	-	-	-
2	E2	4	4	288.6	60.2	2.4	5	*Y9	2	1	197.8	-41.2	-
2	E5	4	4	299.2	58.9	8.9	5	Y10	2	0	-	-	-
2	E6	4	4	267.7	47.0	14.6	6	D1	4	3	338.8	62.6	4.3
2	E9	4	4	285.5	-19.0	33.5	6	D2	4	3	142.2	54.0	28.2
2	*H1	3	3	304.2	70.6	2.8	6	D3	4	2	206.3	63.7	8.8
2	*H4	3	3	344.8	60.4	8.8	6	D4	4	3	052.8	74.2	3.1
2	H5	4	4	296.9	64.1	2.5	6	D5	3	2	129.1	68.6	24.0
2	H6	4	4	270.5	50.0	11.9	6	D6	4	3	084.9	72.1	4.3
2	H8	3	1	059.6	17.0	-	6	D7	4	3	043.7	52.2	29.6
2	H13	4	3	301.9	56.3	3.9	6	D8	4	4	331.2	-08.4	12.5
2	H14	4	3	318.3	59.5	21.2	6	D11	4	3	315.8	18.4	34.8
2	H15	4	1	280.1	59.5	-	6	D12	4	3	180.1	65.2	31.1
2	H16	4	4	282.7	62.9	4.6	6	D13	4	4	069.4	83.2	4.7
2	H17	4	4	276.9	59.7	2.8	6	D14	4	3	049.0	67.1	5.9
2	H18	4	4	287.6	56.6	18.3	6	D15	4	4	347.9	84.7	8.2
2	H19	4	4	323.0	23.9	30.4	6	*R4	3	1	358.7	77.4	-
2	H25	4	4	084.2	-59.3	28.9	6	R10	4	3	065.0	69.3	2.6
2	H26	4	4	343.0	50.7	7.1	6	R13	4	3	013.3	83.1	4.1
2	H27	4	3	342.4	52.8	11.6	7	*M3	3	0	-	-	-
2	H28	4	4	302.8	59.1	5.1	7	*M4	3	1	149.4	67.6	-
2	H29	4	3	297.5	52.3	3.6	8	*B7	3	0	-	-	-
2	H30	4	4	345.8	64.2	6.1	8	*B8	3	0	-	-	-
2	P2	4	4	249.1	-61.4	32.5	9	*G3	3	0	-	-	-
2	P3	4	1	277.0	12.6	-	9	*G4	3	1	098.8	75.2	-
3	*T1	3	0	-	-	-	9	*W6	3	0	-	-	-
3	*T4	3	2	214.8	75.6	7.9							

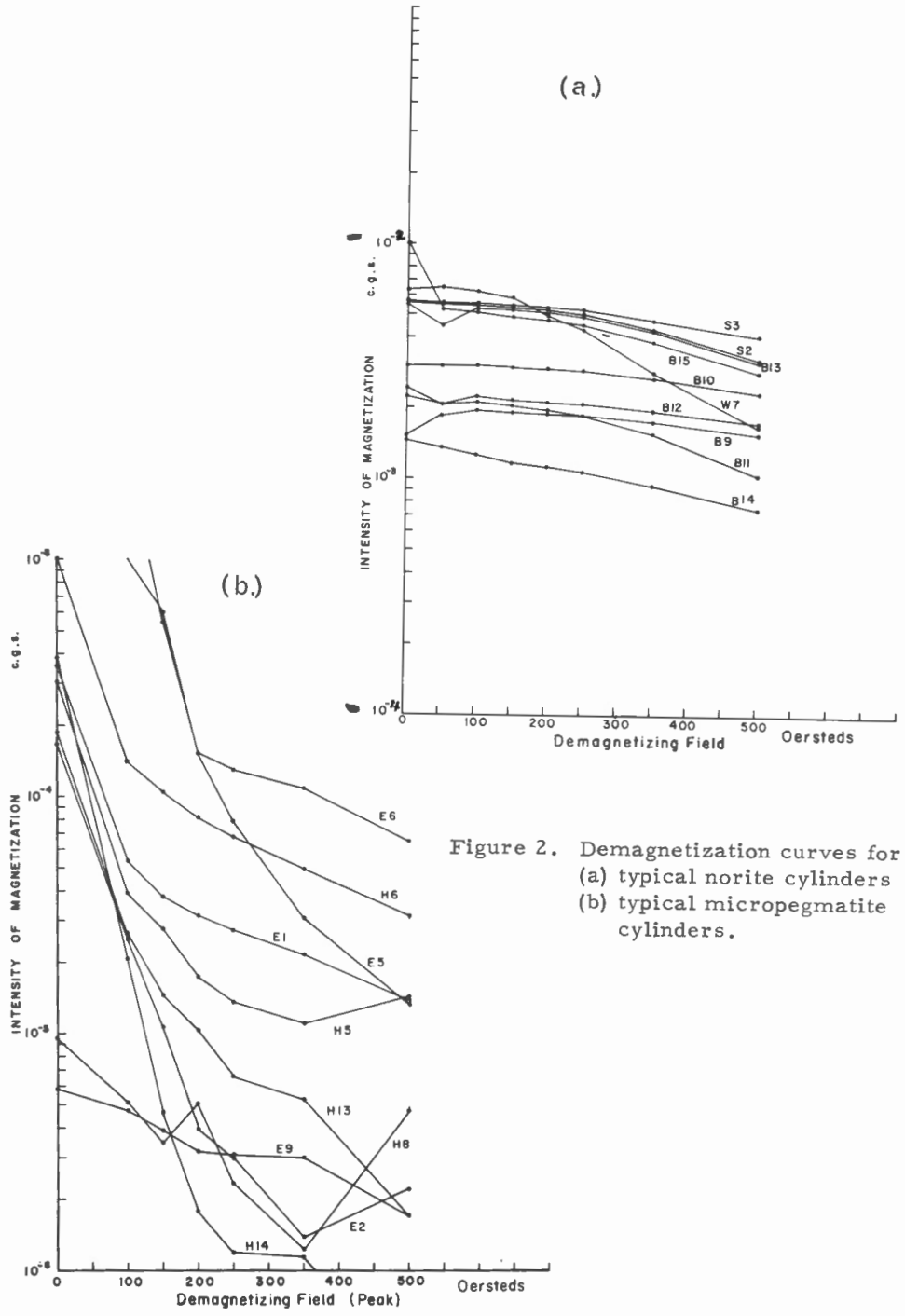


Figure 2. Demagnetization curves for
(a) typical norite cylinders
(b) typical micropegmatite cylinders.

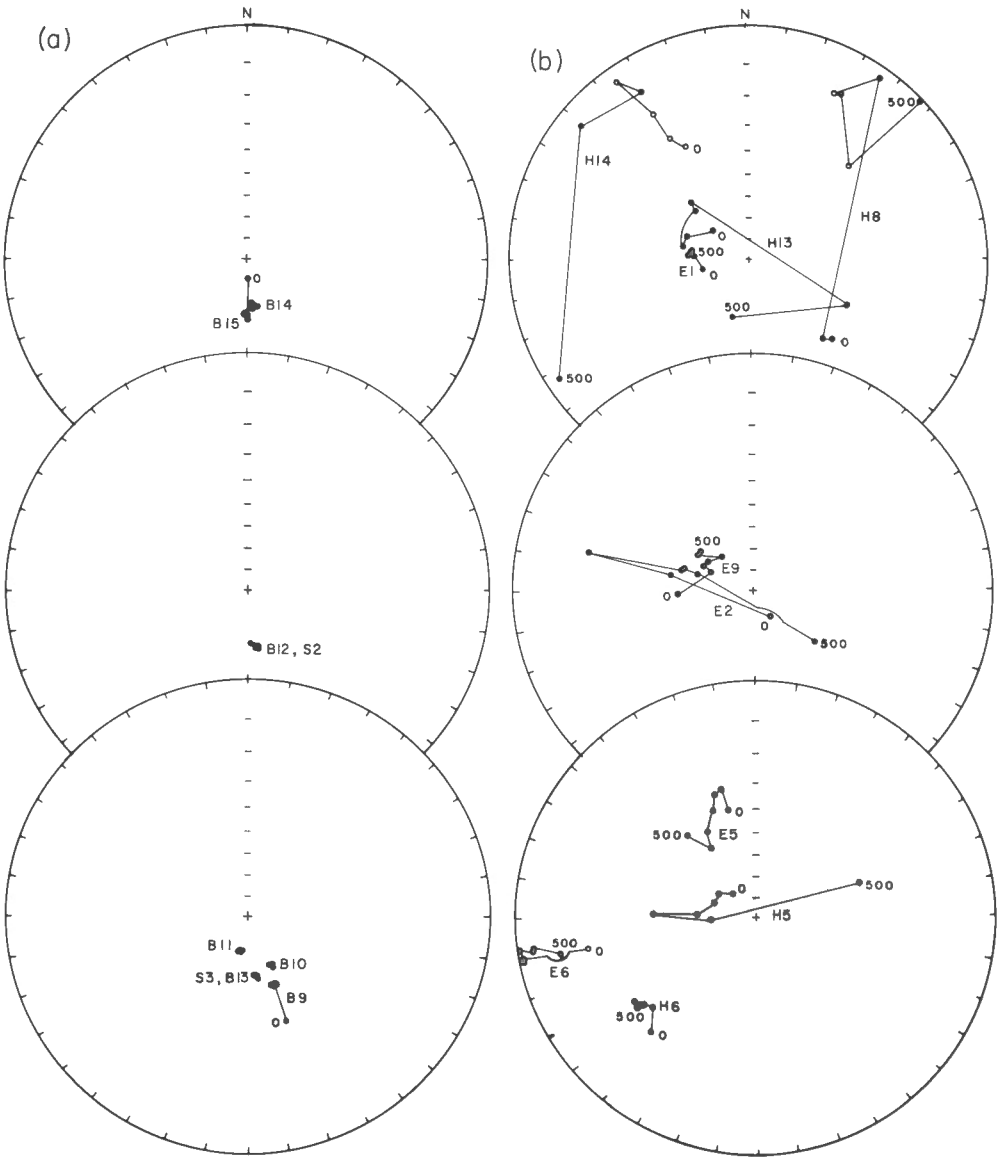


Figure 3. Stereographic projections of successive magnetization directions (joined by lines) of the (a) norite and (b) the micropegmatite cylinders dealt with in Figure 2. NRM directions are opposite 0 and directions after treatment in 500 oe peak field are opposite 500; full (opened) dots refer to directions pointing downward (upward).

RESULTS

The mean direction of magnetization of each 'cleaned' core was taken as that of the bisectrix of the magnetization vectors obtained for the two cylinders cut from it. The angle θ between these two vectors was used as a measure of the relative within-core magnetization homogeneity.

Histograms of θ for both rock types (Figs. 4a, 4b) indicate that in general the norite is appreciably more uniformly magnetized than the micropegmatite. Since, as shown earlier, the magnetization of the norite is also more stable, the paleomagnetic information available from the norite portion of the irruptive can be considered generally more reliable than that obtained from the micropegmatite specimens.

Setting 10 degrees as the maximum standard error ($\delta_{\overline{c}}$) for the magnetization direction of a reliable core, cores for which θ exceeded 20 degrees were assumed to yield unreliable paleomagnetic information since the angular standard error is defined in the present case by

$$\delta_{\overline{c}} = \left[2 - 2\cos\left(\frac{\theta}{2}\right) \right]^{1/2} \quad \text{radians}*$$

The directions of magnetization of the cores considered reliable were represented by unit length vectors and they were grouped by sites. The direction of their resultant at each site will be referred to as the site mean; its standard error was computed on the basis of the relation

$$\delta_{\overline{R}} = \left[\frac{2(n-R)}{n(n-1)} \right]^{1/2} \quad \text{radians}$$

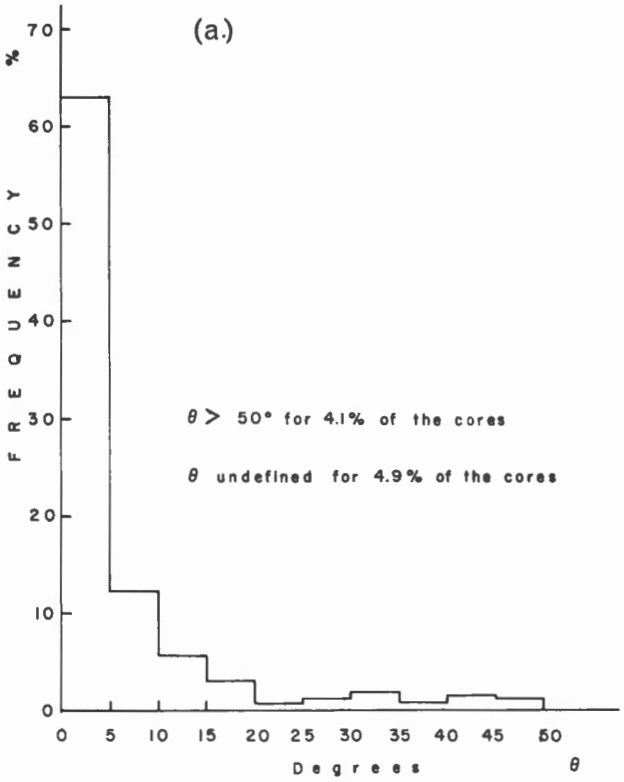
where n is the number of reliable directions and R is the modulus of their resultant at a given site. Tables 1a and 1b indicate that the value of $\delta_{\overline{R}}$ is smaller than 10 degrees for a large number of sites. This suggests that the abnormally high angular dispersion observed at other sites can probably not be attributed to the orientation technique used for the sampling and it may safely be assumed from the remarkably low dispersion at some of the sites that the probable orientation error is less than 2 degrees.

Whenever only one core yielded reliable information at a site, the standard error of the site mean could not be assessed and, owing to this lack of control, the mean direction obtained for this site was considered insufficiently substantiated for further analysis.

Sites for which the value of $\delta_{\overline{R}}$ exceeded 10 degrees normally should be rejected as unreliable if the same standards are adopted for the magnetization homogeneity at the site level as at the core level of sampling. In a few cases, however, a closer examination of the directions obtained for a site indicated that the relatively high value of $\delta_{\overline{R}}$ hinged mainly on one abnormally diverging direction. An example of this is illustrated by the directions obtained for site E6 (Table 2). After elimination of the third vector in the

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* This is essentially a particular case of the next equation where $n = 2$ and $R = 2\cos\left(\frac{\theta}{2}\right)$.



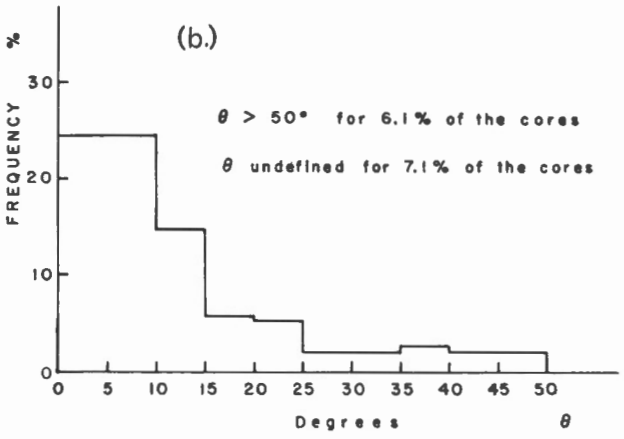


Figure 4. Histograms of θ for
(a) norite cores and
(b) micropegmatite cores.

group E6-3, the statistics (D' = 270.9, I' = 59.9, $\delta_{\frac{1}{R}}$ = 8.2) were computed for the 3 vectors and the odd vector was found to diverge by 54 degrees from the 'adjusted mean direction' (D', I'). As this angle is greater than $(3\sqrt{(n-1)})\ \delta_{\frac{1}{R}}$ (i.e. 3 standard deviations), it can be shown that the probability that vector E6-3 belongs to the same population as the other three vectors is less than 0.002. Accordingly, the 'adjusted' statistics D', I' and $\delta_{\frac{1}{R}}$ were considered more reliable than those listed in Table 1b for site E6. Similar adjustments were only possible for 6 other sites and the corresponding 'adjusted' statistics are listed in Table 3. The data obtained from all other sites for which $\delta_{\bar{R}}$ exceeded 10 degrees were not considered for further analysis. The data considered reliable so far are summarized in Tables 1a, 1b and 3, opposite underlined site numbers.

Table 2: Core Mean Directions Obtained for Site E6

Core No.	θ deg	D deg	I deg	$\delta_{\bar{R}}$ deg
E6-1	4.9	295.3	54.9	-
E6-2	2.5	246.8	66.4	-
E6-3	6.2	263.0	5.9	-
E6-4	3.4	263.6	53.9	-
Mean	-	267.7	47.0	14.6

θ, angle between the directions of magnetization of the two cylinders representative of a core; D, I, declination and inclination of core mean direction of magnetization; $\delta_{\bar{R}}$, site mean standard error.

Table 3: 'Adjusted' Mean Directions of 6 Sites

Site No.	N	n'	D' deg	I' deg	$\delta_{\frac{1}{R}}$ deg	ψ deg
D2	4	2	132.0	79.8	5.4	87
D7	4	2	82.8	74.0	8.2	104
D12	4	2	309.2	85.4	5.9	97
E6	4	3	270.9	59.9	8.2	54
H6	4	3	287.5	53.9	4.6	47
H14	4	2	278.3	58.5	5.9	54
H27	4	2	337.2	42.2	5.4	32

N, total number of cores sampled; n', number of cores left after elimination of one erratic core; D', I' and $\delta_{\frac{1}{R}}$, declination, inclination and standard error of site mean after elimination of one erratic core; ψ, angle between direction of magnetization of erratic core and that defined by D', I'.

No formal evidence is presented here concerning the process by which the rocks of the Sudbury Irruptive were magnetized and such evidence will have to be provided by further work. Nevertheless, the remarkable magnetic stability of the norite and of some of the micropegmatite, leads one to believe that the rocks acquired their present stable magnetization either

through a chemical or a thermoremanent process. The high coherency of the data within a large part of the irruptive would, on the other hand, require that the temperature, pressure, chemical, and atmospheric conditions were uniform over wide areas to be compatible with the hypothesis of chemical remagnetization. This is not impossible but in the light of present knowledge, the hypothesis of primary thermoremanent magnetization appears more likely because it does not imply any of these special conditions. In the following analysis, the basic assumption of thermoremanent magnetization is provisionally made.

STATISTICAL ANALYSIS

After elimination of all data obtained for those sites where either $\delta_{\overline{R}}$ or $\delta_{\frac{1}{R}}$ exceeded 10 degrees or were not defined, the directions of magnetization of the reliable cores were grouped by sites within each sector and the variance-ratio test was used to establish whether the site mean directions were significantly distinct. The test indicates that invariably the site means are significantly distinct because, as shown in Table 4, the ratio of the between-site to the within-site angular variances δ_b^2/δ_w^2 is appreciably greater than the corresponding theoretical statistic F at the .05 probability level. Thus within each sector, the paleomagnetic field secular variation and local anomalies, the magnetization inhomogeneities, and the tectonic adjustments between sites, may be considered as more important factors of angular dispersion than the sampling and measurement errors or the between-core magnetization inhomogeneities within sites. It may therefore be estimated that the within-site factors of angular dispersion are averaged out in the site means and it seems critical to use the latter in preference to the individual core mean directions to estimate the scope of the angular secular variation of the paleomagnetic field.

Of particular interest in Table 4 are the statistics obtained for sector 6: whereas the δ_b^2/δ_w^2 ratio is very high for the norite data, it is among the lowest for the micropegmatite data. The within-site angular variance δ_w^2 of the norite data is almost one third that of the micropegmatite data whereas the between-site variances are in inverse proportion. These statistics are fully compatible with the paleomagnetic observation that, within a site, the norite is generally more homogeneously magnetized than the micropegmatite, and with the geological observation (Fig. 1) that the norite portion of sector 6 appears to have been disturbed to a greater degree than the micropegmatite and that the latter was formed after a first period of tectonic adjustment in the norite.

The 20 site means obtained from the norite part of sector 2 were grouped as a sample and compared to the sample formed by the 16 micropegmatite site means of the same sector. A variance-ratio test indicated (Table 5) that the mean directions of the two samples differ significantly at the .05 probability level but the evidence provided by the test is marginal and the age difference between the two rock types may in fact be as little as a few thousand years. The variance-ratio test applied to the two sets of data obtained for sector 6 would suggest a more appreciable age difference but, for reasons given earlier, these data are not acceptable as valid evidence in this context.

Table 4: Angular Variances and Standard Error of the Mean within Sectors

S	n	B	D deg	I deg	δ^2 rad^2	δ_b^2 rad^2	δ_w^2 rad^2	δ_b^2/δ_w^2	$F_{.05}$	$\delta_{\bar{m}}$ deg
N-1	2	1	236.5	54.5	.00406	-	-	-	-	-
N-2	78	20	292.9	64.9	.02033	.06319	.00638	9.9	1.5	1.6
N-3	7	3	297.7	70.3	.05659	.12337	.02359	5.2	3.8	7.6
N-4	0	0	-	-	-	-	-	-	-	-
N-5	4	2	265.6	86.7	.13024	.32873	.03235	10.1	6.4	16.4
N-6	49	16	66.7	45.0	.21442	.66767	.01105	60.4	1.5	6.7
N-7	3	1	193.8	45.1	.05767	-	-	-	-	-
N-8	56	16	172.5	63.6	.02542	.07559	.00678	11.1	1.5	2.1
N-9	12	4	166.2	62.6	.05502	.18531	.00630	29.4	2.7	7.1
M-1	2	1	240.3	46.2	.02648	-	-	-	-	-
M-2	54	16	303.8	61.2	.07968	.20111	.03286	6.1	1.6	3.5
M-3	2	1	214.8	75.6	.03769	-	-	-	-	-
M-4	0	0	-	-	-	-	-	-	-	-
M-5	0	0	-	-	-	-	-	-	-	-
M-6	34	12	49.8	81.6	.08350	.19362	.02928	6.6	1.7	4.3
M-7	0	0	-	-	-	-	-	-	-	-
M-8	0	0	-	-	-	-	-	-	-	-
M-9	0	0	-	-	-	-	-	-	-	-

S, sector number (e.g. N-1 stands for norite data from sector No. 1); n, total number of reliable cores; B, number of sites; D, I, declination and inclination of resultant of 'weighted' site means or of the n directions; δ^2, angular variance of sector; δ_b^2, δ_w^2, between-site and within-site angular variances; $F_{.05}$, theoretical statistic $F_{2(B-1),\ 2(n-B),\ .05}$; $\delta_{\bar{m}} = \delta_b/\sqrt{n}$, angular standard error of 'weighted' site means resultant.

Table 5: Within- and between-rock type (-sector) Angular Variances

Sample 1	Sample 2	B	δ_b^2 rad^2	δ_w^2 rad^2	δ_b^2/δ_w^2	$F_{.05}$
N-2	M-2	36	.13949	.03996	3.5	3.1
N-6	M-6	28	2.63225	.16644	15.8	3.2
N-2	N-8	36	(5.11791)	(.02339)	218.8	3.1

δ_b^2, δ_w^2, between-sample and within-sample angular variances of normalized site means; $F_{.05}$, Theoretical statistic $F_{2,2(B-2),.05}$

Assuming that the sampling and measurement errors are averaged out in the site means, it appears that the angular dispersion of the latter within a profile across the structure can be attributed to relative displacements or to secondary magnetization of the sites or to the paleomagnetic field secular variation and local anomalies. It may sometimes be fairly evident from field observation that the surface rocks along a profile have not been tilted differentially but it remains difficult to decide which of (1) the paleomagnetic field secular variation and local distortions and (2) the secondary magnetization is the more important cause of angular dispersion of the site mean directions. Therefore one can only estimate the maximum value of the paleomagnetic field secular variation by ignoring the effects of local paleomagnetic field anomalies and secondary magnetization. On the basis of this assumption, computations were made of the angular standard deviation of site means chosen along 3 profiles in sectors 2 and 8. The site means chosen and the standard deviation for each profile are listed in Table 6. The values obtained for δ varied between 5.4 and 14.8 degrees and the profile mean directions all have an inclination about 60 degrees. On the basis of curves proposed by Cox (1962) for the angular dispersion of the earth's field over a period of 10,000 years or more, it could be suggested that the cooling of either the norite or the micropegmatite rims was completed over that long a period if the values of δ were of the order of 15 degrees. The value of δ for at least one of the profiles indicates however that the cooling of at least the norite phase of the irruptive either took place within an appreciably shorter period of time or that the earth's field was more stable at that time than it is today.

Table 6: Angular Standard Deviations of Profile Samples

Profile No.	Sites	δ degrees
1	H-2, 3, 7, 11, 12, 20, 21, 22	5.4
2	H-1, 4, 5, 26, 27, 28, 29, 30	14.8
3	B-2, 3, 4, 5, 6, 11, 14	13.0

δ, angular standard deviation of 'normalized' site means

Finally, the variance-ratio test was used to verify on a statistical basis that the mean direction of magnetization obtained for sector 2 is significantly different from that obtained from sector 8. One might interject that

Table 7a: Site Mean Directions (Sopher's Data); Norite Sites
(for explanation of symbols see Table 1a)

S	s	N	n	D (deg)	I (deg)	$\delta_{\overline{R}}$ (deg)	S	s	N	n	D (deg)	I (deg)	$\delta_{\overline{R}}$ (deg)
1	80	2	2	278.5	75.1	1.4	7	31	2	2	155.4	44.1	18.5
1	83	2	2	263.4	57.0	1.7	8	19	2	2	151.4	64.0	3.0
1	84	2	2	260.5	54.5	1.6	8	20	2	2	151.0	37.0	2.6
1	85	2	2	262.7	63.1	2.7	8	23	2	2	148.8	70.7	6.1
2	3	2	2	285.4	61.5	4.7	8	26	2	2	154.8	64.6	6.8
2	89	2	2	340.9	66.5	6.5	8	14	2	2	173.9	65.5	1.3
2	4	2	2	299.6	63.6	2.1	8	25	2	2	164.0	68.0	1.1
2	5	2	2	311.5	59.4	10.0	8	27	2	2	180.9	73.0	3.7
2	7	2	2	260.3	14.5	11.3	8	41	2	2	150.4	54.2	10.8
3	9	2	2	287.1	62.5	2.1	8	49	2	2	172.6	68.0	4.0
3	8	2	2	294.9	49.2	27.4	9	42	2	2	193.2	61.3	4.2
3	94	2	2	359.6	70.7	3.0	8	18	2	2	183.3	67.5	2.8
4	93	2	2	120.5	-88.6	3.3	9	45	2	2	168.1	63.9	7.9
5	56	2	2	43.4	74.5	2.5	9	46	2	2	150.0	56.1	2.8
6	33	2	2	70.5	43.6	6.6	9	116	2	2	203.5	62.5	0.8
6	59	2	2	75.5	37.5	3.5	9	117	2	2	187.7	62.6	2.8
6	97	2	2	21.2	74.0	15.0	9	109	2	2	169.0	74.4	30.3
6	98	2	2	113.9	74.0	8.4	9	111	2	2	245.0	67.5	0.6
7	29	2	2	189.7	24.2	8.3	9	113	2	2	240.8	70.9	13.7
7	30	2	2	178.4	42.0	2.8	10	60	2	2	274.0	-2.0	21.1

Table 7b: Site Mean Directions (Sopher's Data) Micropegmatite
(for explanation of symbols see Table 1a)

S	s	N	n	D (deg)	I (deg)	$\delta_{\overline{R}}$ (deg)	S	s	N	n	D (deg)	I (deg)	$\delta_{\overline{R}}$ (deg)
1	$\underline{78}$	2	2	252.9	34.9	9.3	4	100	2	2	188.1	20.2	17.8
2	1	2	2	255.3	43.6	18.7	5	108	2	2	286.7	41.2	22.5
3	$\underline{50}$	2	2	295.3	79.9	9.0	5	$\underline{53}$	2	2	331.4	79.5	4.7
3	$\overline{51}$	2	2	258.6	75.8	12.0	6	$\underline{35}$	2	2	34.4	64.0	9.5
3	$\underline{52}$	2	2	305.7	83.4	9.0	6	$\underline{36}$	2	2	66.1	42.7	6.5
3	$\underline{95}$	2	2	346.2	69.4	8.5	6	$\underline{57}$	2	2	73.4	69.7	9.8
3	$\underline{11}$	2	2	302.8	65.0	2.2	6	$\underline{96}$	2	2	54.9	71.1	8.0
4	$\overline{99}$	2	2	236.9	49.5	20.2	8	$\underline{22}$	2	2	318.8	81.4	4.1
4	104	2	2	291.4	47.0	23.4	8	$\underline{115}$	2	2	350.6	58.3	11.5
4	91	2	2	302.5	14.6	31.0	9	$\underline{65}$	2	2	106.7	75.0	6.8

this is obvious from even a superficial examination of Sopher's or Hood's data, but the result of the test in Table 5 is given to illustrate how conclusive an answer the test gives in such an obvious case.

INCORPORATION OF EARLIER DATA

The Sudbury Irruptive paleomagnetic data published earlier cannot all be compatibly blended with the above and it seems reasonable to accept as complementary to the present study only the information satisfying the specifications adopted earlier in this paper. For the data reported by Sopher (op. cit.) it is specified that two oriented samples were collected at each site, that these were oriented in situ with a solar compass and that the mean direction of magnetization of each sample is based on the measurements of two cube specimens, the latter having been cleaned in a demagnetizing field of 280 oe. The magnetization directions reported in Appendix I of Sopher's paper were re-examined in the light of the norms adopted in the present study. Consequently only sites for which two valid directions were reported by Sopher were included in Tables 7a and 7b, where these data are summarized as site means for the norite and micropegmatite samples and where the sector's pattern of Figure 1 was adopted.

It is noted in Tables 7a and 7b that the standard errors of the site means are again generally less than 10 degrees for the norite, whereas the clustering of the individual core means about the micropegmatite site means is generally less accentuated. In accordance with the previously adopted norm the data reported for those sites where $\delta_R > 10$ degrees are not included throughout the rest of this analysis.

The counterpart of Table 4 based on Sopher's data is given in Table 8. A comparison of corresponding entries in the two tables often shows that the sectors mean directions obtained from Sopher's data are roughly parallel to those reported in Table 4 although Sopher's data are generally characterized by a slightly greater angular dispersion. To establish that the two sets of data are in fact equally reliable and that they can complement each other, the following statistical analysis was made.

For each of the entries where δ_m is defined in both tables, the site means were 'normalized' (i.e. represented by unit vectors) and grouped as a sample. The angular variance δ^2 of each sample and the within- and between-sample angular variances were computed for each pair of such samples. The results of these computations (Table 9) were used to verify by variance-ratio test (a) whether the dispersions of the two samples in a pair are significantly distinct and (b) whether the means of the same pair of samples were significantly distinct. The tests show invariably (Table 9) that the two sets of data are not significantly distinct because for each pair of samples, $\delta_b^2/\delta_w^2 < F_{2,\ 2(B_1+B_2-2),\ .05}$ and $\delta_1^2/\delta_2^2 < F_{2(B_1-1),\ 2(B_2-1),\ .05}$. It is therefore legitimate to blend the two sets of data into one, as will be done in the rest of this analysis.

Table 8: Angular Variances and Standard Error of the Mean within Sectors (Sopher's data)
(for explanation of symbols see Table 4)

S	n	B	D	I	δ^2	δ_b^2	δ_w^2	δ_b^2/δ_w^2	$F_{.05}$	$\delta_{\bar{m}}$
			deg	deg	rad^2	rad^2	rad^2			deg
N-1	8	4	264.4	62.5	.02557	.05674	.00221	25.7	3.6	4.8
N-2	8	4	308.0	64.1	.04399	.06885	.02565	2.7	3.6	5.3
N-3	4	2	316.5	70.6	.08161	.23687	.00410	57.7	6.9	13.9
N-4	2	1	120.5	-88.6	.00643	-	-	-	-	-
N-5	2	1	43.4	74.5	.00382	-	-	-	-	-
N-6	6	3	79.0	52.4	.13581	.30360	.02563	11.8	4.5	12.9
N-7	4	2	184.6	33.3	.05620	.12275	.02329	5.2	6.9	10.0
N-8	18	9	165.7	65.0	.04668	.08898	.00920	9.7	2.2	4.0
N-9	12	6	187.9	65.2	.06341	.12784	.00985	13.0	2.8	5.9
M-1	2	1	252.9	34.9	.05266	-	-	-	-	-
M-2	0	0	-	-	-	-	-	-	-	-
M-3	8	4	315.8	75.4	.05048	.07010	.03617	1.9	3.6	5.3
M-4	0	0	-	-	-	-	-	-	-	-
M-5	2	1	331.4	79.5	.01382	-	-	-	-	-
M-6	8	4	50.3	66.3	.15045	.29470	.04459	6.6	3.6	11.0
M-7	0	0	-	-	-	-	-	-	-	-
M-8	2	1	318.8	81.4	.01033	-	-	-	-	-
M-9	2	1	106.7	75.0	.02793	-	-	-	-	-

Table 9: Variance-ratio tests to compare present and Sopher's sets of data

Sample 1	Sample 2	B_1	B_2	δ_1^2 rad^2	δ_2^2 rad^2	δ_1^2/δ_2^2	$F_{.05}$	δ_b^2 rad^2	δ_w^2 rad^2	δ_b^2/δ_w^2	$F'_{.05}$
NS-2	N-2	4	20	.03442	.02068	1.67	2.35	.05409	.02257	2.40	3.21
NS-3	N-3	2	3	.11844	.05879	2.02	6.94	.01765	.07864	0.22	5.14
N-6	NS-6	16	3	.23191	.15180	1.53	5.74	.13199	.22242	0.59	3.28
NS-8	N-8	9	16	.04449	.02686	1.65	1.99	.01442	.03300	0.44	3.20
NS-9	N-9	6	4	.06392	.06177	1.03	4.06	.06991	.06305	1.11	3.63
MS-6	M-6	4	12	.14755	.07715	1.91	2.55	.24145	.09230	2.62	3.34

Sample labelled NS-2 refers to Sopher's norite data for sector No. 2; δ_1^2, δ_2^2, angular variances of sample 1 and 2 made up respectively of the B_1 and B_2 'normalized' site means; $F_{.05}$, theoretical statistic $F_{2(B_1-1),\ 2(B_2-1),\ .05}$; δ_b^2, δ_w^2, between-sample and within-sample angular variances; $F'_{.05}$, theoretical statistic $F_{2,\ 2(B_1+B_2-1),\ .05}$.

Table 10: Summary of reliable paleomagnetic data available for the Sudbury Irruptive

	norite data					micropegmatite data				
S	n	B	D deg	I deg	$\delta_{\overline{m}}$ deg	n	B	D deg	I deg	$\delta_{\overline{m}}$ deg
1	10	5	257.8	61.4	5.0	4	2	247.1	40.8	7.4
2	86	24	294.3	64.9	1.6	54	16	303.8	61.2	3.5
3	11	5	304.4	70.6	6.2	10	5	301.4	78.5	6.1
4	2	1	120.5	-88.6	-	0	0	-	-	-
5	6	3	22.0	86.0	11.2	2	1	331.4	79.5	-
6	55	19	62.5	46.0	6.0	42	16	50.0	78.8	4.2
7	7	3	188.2	38.4	7.4	0	0	-	-	-
8	74	25	170.9	64.0	1.9	2	1	318.8	81.4	-
9	24	10	176.5	64.3	4.6	2	1	106.7	75.0	-
10	0	0	-	-	-	0	0	-	-	-

S, sector number; n, (B), total number of reliable cores (sites) included in both Sopher's and present data; D, I, declination and inclination of 'weighted' site means resultant; $\delta_{\overline{m}}$, angular standard error of 'weighted' site means resultant.

Table 11: Variance-ratio test to compare norite and micropegmatite samples of sectors 2, 3 and 6 (for explanation of symbols see Table 9)

Sample 1	Sample 2	B_1	B_2	δ_1^2 rad^2	δ_2^2 rad^2	δ_1^2/δ_2^2	$F_{.05}$	δ_b^2 rad^2	δ_w^2 rad^2	δ_b^2/δ_w^2	$F'_{.05}$
M-2	N-2	16	24	.06439	.02391	2.69	1.71	.11102	.03989	2.78	3.12
N-3	M-3	5	5	.06328	.05619	1.12	3.44	.05336	.05974	0.89	3.63
N-6	M-6	19	16	.21695	.10143	2.14	1.81	2.49124	.16444	15.15	3.14

DISCUSSION

The paleomagnetic data considered above are summarized in Table 10. It is obvious from this summary that all sector means are not equally well substantiated and that additional sampling is required in some of the sectors before a comprehensive geological interpretation of the irruptive can be deduced from the evidence provided by its paleomagnetism. For instance, the lack of information from sectors 4 and 10 is evident and sectors 5 and 7 should be sampled more extensively before an attempt is made to explain their structural relationships in the light of paleomagnetism.

Unlike the micropegmatite means of all other sectors which are inadequately substantiated those of sectors 2, 3 and 6 stand comparison with their norite counterparts. The angular variances of the two samples (i. e. the norite and the micropegmatite 'normalized' site means respectively) were computed for each of these three sectors and the δ_1^2/δ_2^2 ratios were compared to the statistic $F_{2(B_1-1),\ 2(B_2-1),\ .05}$. Table 11 (columns 7 and 8) indicates that only the two samples of sector 3 have equivalent dispersions although the between-sample to the within-sample variance ratios (column 12, Table 11) indicates that the two sector means are not diverging significantly in sectors 2 and 3. It is noted however that the result obtained for sector 2 is marginal, much like the contradicting indication obtained by the same test, prior to the incorporation of Sopher's data (Table 5). When applied to the two samples representative of sector 3, the test suggests more strongly that the micropegmatite and the norite are of the same paleomagnetic age.

An explanation for the obvious divergence of the two means obtained for sector 6 was given earlier but an examination of Table 10 suggests another possibility. The micropegmatite means of sectors 3 and 6 have roughly the same declinations as their norite counterparts but they are appreciably steeper, similar to the other micropegmatite sector means represented by only one site mean. This suggests that a steeply dipping component of chemical magnetization due to alteration may have been differentially but systematically added to the primary thermoremanent component of magnetization in the micropegmatite. This hypothesis would be compatible with the generally greater magnetization inhomogeneity observed within individual cores and sites of the micropegmatite. On the other hand, the hypothesis is not incompatible with the suggestion made above that the norite has been disturbed prior to the formation of the micropegmatite because tectonic displacement and chemical alteration may both have contributed to the 30 degree divergence between the two means obtained for sector 6. One obvious objection to this interpretation is the shallow inclination of the sector 1 micropegmatite mean, but more data are needed from this portion of the irruptive to verify the validity of this objection.

It would be superfluous to make further use of the statistical methods to demonstrate that the sufficiently substantiated norite sector means in the North Range diverge significantly from their counterparts in the South Range. It is interesting to note on the other hand that the means obtained for sectors 1, 2 and 3 diverge appreciably from one another whereas the norite means of sectors 8 and 9 (which actually represent a larger portion of the irruptive) are not significantly different. This would suggest that the North Range is made up of differentially rotated blocks whereas the portion of the South Range in sectors 8 and 9 behaved rather as a unit.

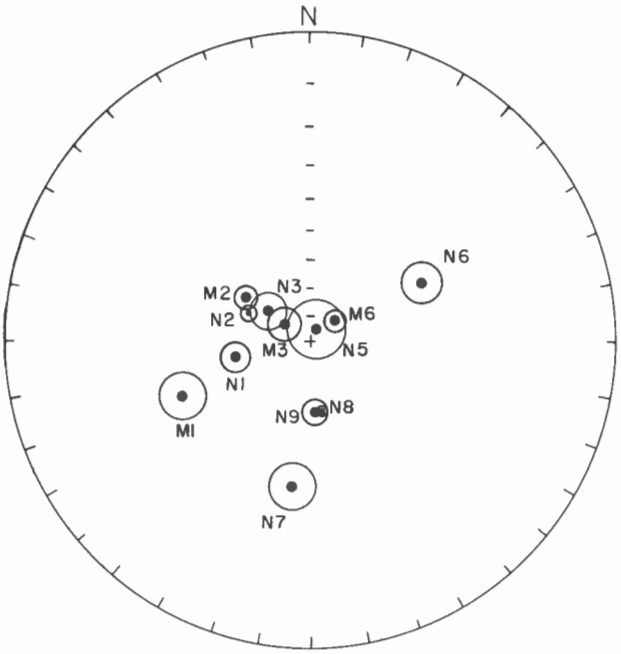


Figure 5. Stereographic projection of reliable sector mean directions. Code letters N and M refer to norite and micropegmatite data respectively. Radii of circles about sectors means represent their standard error and they are equal to $\alpha'_{.95}/\sqrt{3}$.

In view of the many uncertainties concerning the geological history of the nickel irruptive it appears premature to calculate pole positions on the basis of the sector means reported in Table 10. Because the original attitude of the irruptive cannot be visualized yet with any kind of assurance in the light of the available data, such pole positions would only add unreliable kinks to the 'pre-Cambrian polar wander curve'.

Figure 5 sums up as a stereoplot the sector means accepted in this paper as reliable. There is no doubt that considerably more information is required from large portions of the irruptive before paleomagnetism can be used as a safe guide in unravelling its structural history and this is why these results are considered as preliminary.

SUMMARY AND CONCLUSIONS

Although this study provides no formal evidence concerning the process of magnetization of the rocks dealt with, the high magnetic stability and coherency of the norite within sectors of the irruptive suggest that it acquired the major component of its remanent magnetization at the time of its original cooling. The considerably lower magnetic homogeneity of the micropegmatite

suggests, on the other hand, that an appreciable percentage of its remanent magnetization may in fact reside in minerals formed long after the original cooling of this phase of the irruptive.

As the norite part of the irruptive proved to be a more reliable carrier of paleomagnetism than the micropegmatite, it appears that it is more apt than the latter to help in the clarification of some of the present ambiguities concerning the structural history of the irruptive. The micropegmatite data obtained so far were useful however in showing that the two phases of the irruptive are probably contemporaneous.

Since the angular standard deviation of a set of paleomagnetic directions can only be used to evaluate the maximum scope of the paleomagnetic field secular variation during a specific geological period, the relatively small angular standard deviations observed at least in the norite indicate that either this phase of the irruptive was an event of short duration or that the paleomagnetic field was more stable at that time than it is today.

All available data on the paleomagnetism of the Sudbury Irruptive indicate that the mean directions of magnetization in the North and South Ranges diverge significantly by about 40 degrees, in conformity with the previous observations of Hood (1961) and Sopher (1963).

A large proportion of the paleomagnetic data previously reported by Sopher satisfy the norms imposed on the data reported in the present study and can legitimately be combined with them. None of the previous paleomagnetic interpretations can yet be disputed or corroborated in the light of the above analysis but their basis and the planning of future work to test them is undoubtedly established more firmly.

REFERENCES

Cox, A.
1962: Analysis of the present geomagnetic field for comparison with paleomagnetic results; J. Geom. and Geoel., vol. 13, p. 101.

Fairbairn, H.W., G. Faure, W.H. Pinson Jr., P.M. Hurley
1968: Rb-Sr whole-rock age of the Sudbury lopolith and basin sediments; Can. J. Earth Sci., vol. 3, p. 707.

Hood, P.J.
1961: Paleomagnetic study of the Sudbury Basin; J. Geophys. Res., vol. 66, p. 1235.

Larochelle, A.
1968: L'application de la statistique au paléomagnétisme; Geol. Surv. Can., Paper 68-59.

Larochelle, A. and Black, R.F.
1965: The design and testing of an alternating field demagnetizing apparatus; Can. J. Earth Sci., vol. 2, p. 684.

Larochelle, A. and Christie, K. W.
1967: An automatic 3-magnet or biastatic magnetometer; Geol. Surv. Can., Paper 67-28.

Sopher, S. R.
1963: Paleomagnetic study of the Sudbury Irruptive; Geol. Surv. Can., Bull. 90.

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DATE - Thursday, April 19, 1979
jeudi, 19 avril, 1979

TIME/HEURE - 9:00

LOACTION/LIEU - Assembly Room — Salle de Rencontre
Geophysical Laboratory — Laboratoire géophysique
Observatory Crescent. — Place de l'Observatoire

...coffee afterwards
...café servi après

ABSTRACT

A detailed investigation of approximately 490 specimens of Sudbury Norite has isolated seven distinct remanences that co-exist in this single rock unit. Complex thermal demagnetisation behaviour indicates that most of these remanences were at least in part acquired by a chemical magnetisation process. It was not possible to erect a chronology of remanence acquisition events based purely on unblocking temperature relationships. Rather a chronology was established by comparing the metamorphic grade of differing specimens, the areal distribution of a particular remanence phase, and by the relationship of the remanence direction to the local geological structure. The results, therefore, indicate that at least six-post intrusional events have been of sufficient local severity to leave a lasting imprint on the magnetic record of the Sudbury Norite. Recognition of the sequence of these remanence recording events permits the development of a tectono-metamorphic history of the Sudbury Norite. The absolute timing of these events is unknown.

At the time of formation (N1) the Sudbury Norite occupied an open syncline (limbs dipping approximately 20°) plunging gently to the West-South-West. At some time after this initial structure was compressed (F1) into a crudely symmetrical basin, all limbs dipping between 40-50°. The completion of this tectonic event is signified by the regional metamorphic event recorded as N2 on the South, East and West Ranges. On the South Range where N2 is dominant, the metamorhpism was nodal and some zones of N1 survived with no trace of the development of N2. Next the East and South Ranges were further tilted by a second deformational event (F2) which affected only these limbs. After this second deformation N3 and N4 were imprinted on the Norite. The mechanism that gave rise to these magnetisations is uncertain. However the dominant distribution of N3 in the uppermost parts of the Norite suggests that it is somehow related to development of the oxide-rich gabbro or the transition zone. Subsequent to the acquisition of N3 and N4, the South Range was rotated (F3) en block relative to the North Range. The last event (N7) recorded by the basin was a thermal overprinting at 1000 Ma, related to the developing Grenville Front to the South. The significance of N6 is not presently understood, and it should become clear once further investigations in the Sudbury Basin have been completed.

Two phases of remanence carrying sulphide generation have been recognized. The time of formation of the N5 phase is uncertain. It now carries a direction which reflects the final post F1 cooling of the South and East Range metamorphic event. The sulphide phase associated with N4 indicates that it postdates the F1 and F2 tectonic events. Zonation of the relative dominance of N4 provides a guide to regions of economic interest.